

Lecture 10: Thermal Properties of Matter – Tutorial Sheet

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Use the Master Procedure: identify the system and whether the amount of gas is fixed, sketch (the path on a pV diagram, the heating curve or the molecule and wall), convert units (kelvin, absolute pressure, m^3 , kg/mol), predict, select the model, execute, evaluate. Problems A1 and A2 are solved in full; final answers to the others are at the end. Useful values: $R = 8.314 \text{ J}/(\text{mol K})$, $k = 1.381 \times 10^{-23} \text{ J/K}$, $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$, $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$; $c_{\text{water}} = 4186$, $c_{\text{ice}} = 2100$, $c_{\text{steam}} = 2010 \text{ J}/(\text{kg K})$; $L_f = 3.34 \times 10^5$, $L_v = 2.256 \times 10^6 \text{ J/kg}$ (water). Molar masses (g/mol): H_2 2.02, N_2 28.0, O_2 32.0, air 28.8, Ar 39.9.

A. Worked examples

A1. A gas cylinder in the Sun

A cylinder of volume 12.0 L holds oxygen at a *gauge* pressure of 20.0 MPa and a temperature of 20°C. (a) How many moles of oxygen does it hold? (b) What volume would this oxygen occupy at 1.00 atm and 20°C? (c) The cylinder is left in the Sun and warms to 50°C. Find the new gauge pressure.

Solution. The gas laws need the absolute pressure: $p_1 = 20.0 \times 10^6 + 1.013 \times 10^5 = 2.010 \times 10^7 \text{ Pa}$, $T_1 = 293.15 \text{ K}$, $V = 12.0 \times 10^{-3} \text{ m}^3$. (a) $n = \frac{p_1 V}{RT_1} = \frac{(2.010 \times 10^7)(12.0 \times 10^{-3})}{(8.314)(293.15)} = 99.0 \text{ mol}$. (b) At 1.00 atm, $V = \frac{nRT}{p} = \frac{(98.97)(8.314)(293.15)}{1.013 \times 10^5} = 2.38 \text{ m}^3 = 2.38 \times 10^3 \text{ L}$. *Check:* the pressure ratio is 198, and $198 \times 12.0 \text{ L} = 2.38 \times 10^3 \text{ L}$. (c) The volume is fixed (neglect the expansion of the steel), so $p_2 = p_1 \frac{T_2}{T_1} = (2.010 \times 10^7) \frac{323.15}{293.15} = 2.216 \times 10^7 \text{ Pa}$ absolute, and $p_{2,\text{gauge}} = 2.216 \times 10^7 - 1.0 \times 10^5 = 2.21 \times 10^7 \text{ Pa} = 22.1 \text{ MPa}$. *Check:* the absolute pressure rises by 10.2%, the same as the kelvin temperature. Here p_{atm} is only 0.5% of p_1 , so the gauge ratio would be nearly right; for a tyre it is not (problem B1).

A2. Ice in warm water

0.300 kg of ice at -20°C is dropped into 0.500 kg of water at 40°C in an insulated container. Find the final state.

Solution. *Decide the final state first.* Heat available from the water cooling to 0°C : $0.500(4186)(40) = 8.37 \times 10^4 \text{ J}$. Heat needed to warm the ice to 0°C : $0.300(2100)(20) = 1.26 \times 10^4 \text{ J}$; to melt all of it: $0.300(3.34 \times 10^5) = 1.002 \times 10^5 \text{ J}$; total $1.128 \times 10^5 \text{ J}$, which exceeds the available $8.37 \times 10^4 \text{ J}$. So not all the ice melts and the final temperature is 0°C . *Mass melted:* $m_m = \frac{8.372 \times 10^4 - 1.26 \times 10^4}{3.34 \times 10^5} = 0.213 \text{ kg}$. Ice remaining: $0.300 - 0.213 = 0.087 \text{ kg}$; water: $0.500 + 0.213 = 0.713 \text{ kg}$. *Check:* heat given by the water = $8.37 \times 10^4 \text{ J}$; heat taken by the ice = $1.26 \times 10^4 + (0.2129)(3.34 \times 10^5) = 8.37 \times 10^4 \text{ J}$. ✓

B. Practice

B1. A car tyre

A car tyre has a gauge pressure of 220 kPa at 10°C. After a long drive its temperature is 50°C. Assume the volume and the amount of air are constant and $p_{\text{atm}} = 101.3 \text{ kPa}$. (a) Find the new gauge pressure. (b) By what percentage does the rms speed of the molecules increase? (c) Explain why applying the ratio to the gauge pressure gives the wrong answer.

B2. Air in a classroom

A classroom measures $4.0 \text{ m} \times 5.0 \text{ m} \times 2.5 \text{ m}$ and is at 20°C and 1.00 atm. Find the number of moles, the number of molecules and the mass of air in the room.

B3. Density

Find the density of air and of helium at 25°C and 1.00 atm (use $\rho = pM/RT$). Which fraction of the weight of the air displaced by 1.00 m^3 of helium is the weight of the helium?

B4. A rising bubble

An air bubble of volume 2.0 cm^3 forms in a lake at a depth where the absolute pressure is 3.0 atm and the temperature is 10°C. It rises to the surface, where the pressure is 1.0 atm and the temperature is 20°C. Find its volume at the surface.

B5. States on a pV diagram

A fixed amount, 0.40 mol, of an ideal gas can be in three states: a ($1.0 \times 10^5 \text{ Pa}$, 8.0 L), b ($2.0 \times 10^5 \text{ Pa}$, 5.0 L), c ($4.0 \times 10^5 \text{ Pa}$, 2.0 L). (a) Sketch the three points and the isotherms through them. (b) Which state is hottest, and are any two tied? (c) Find the three temperatures.

B6. A full heating curve

Find the heat needed to take 0.500 kg of ice at -20°C to steam at 120°C at normal pressure. Sketch the heating curve and state which stage takes the most heat. How long does a 2.0 kW heater take if all its energy is absorbed?

B7. Cooling a drink

0.250 kg of juice (treat it as water) at 22°C is cooled by adding ice at 0°C . What mass of ice must be added to bring the final temperature to 4.0°C ?

B8. Steam and ice

0.100 kg of steam at 100°C is passed into 0.500 kg of ice at 0°C in an insulated container. Decide the final state and find the final temperature.

B9. Reading the water phase diagram

The triple point of water is at 273.16 K and 610 Pa. (a) Can liquid water exist at 400 Pa? What happens to ice heated at that pressure? (b) Why does water boil at a lower temperature on a mountain? (c) Why does a pressure cooker cook food faster?

B10. Speeds of molecules

Find the rms speed and the average translational kinetic energy per molecule of (a) oxygen and (b) hydrogen at 300 K. By what factor is the hydrogen faster? Explain this factor.

B11. A target speed

At what temperature is the rms speed of nitrogen molecules 1000 m/s? By what factor would the temperature have to change to double the rms speed of any gas?

B12. Are the molecules point-like?

At 25°C and 1.00 atm, find the number of molecules per cubic metre and the average distance between neighbouring molecules, estimated as $(V/N)^{1/3}$. Compare with a molecular diameter of about 0.3 nm and comment on the assumption that molecules are point-like.

B13. Challenge: the method is not given

2.0 mol of argon sits in a rigid sealed 20.0 L vessel at 27°C. It is heated until the rms speed of the atoms is doubled. (a) Find the initial pressure. (b) Find the final temperature and pressure. (c) Find the initial and final total translational kinetic energy, and the ratio. (d) Show that $\Delta p = \frac{2}{3} \Delta K_{\text{tot}}/V$ for this process.

Exam-style practice

Ten five-option items in the style of the course examination, **suggested time: 3 minutes per item**. For each item sketch first, predict the sign or size, then calculate (IDENTIFY, SET UP, EXECUTE, EVALUATE). Choose one answer, A to E. Items marked as sharing information are linked. Answers and brief solutions are at the end of this sheet. Take $g = 9.80 \text{ m/s}^2$.

E1. What volume is occupied by 2.5 mol of an ideal gas at 1.00 atm ($1.013 \times 10^5 \text{ Pa}$) and 27°C?

- (A) 5.5 L (B) 6.2 L (C) 56.0 L (D) 61.6 L (E) 615.7 L

E2. A fixed amount of ideal gas at 1.00 atm, 2.0 L and 300 K is changed to 3.0 L and 360 K. The final pressure is

- (A) 0.56 atm
(B) 0.80 atm
(C) 1.00 atm
(D) 1.25 atm
(E) 1.80 atm

Items E3 and E4 share this information: Nitrogen ($M = 28.0 \text{ g/mol}$) is treated as an ideal gas at 300 K.

E3. The root-mean-square speed of the molecules is

- (A) 298 m/s
- (B) 422 m/s
- (C) 476 m/s
- (D) 517 m/s
- (E) 2.67×10^5 m/s

E4. At what temperature would the rms speed be twice the value in the previous item?

- (A) 424 K (B) 600 K (C) 900 K (D) 1200 K (E) 2400 K

E5. Which of the following statements about an ideal gas is/are true?

- I. At a given temperature every gas has the same average translational kinetic energy per molecule, $\frac{3}{2}k_B T$.
- II. At the same temperature, molecules of a lighter gas have a larger rms speed.
- III. Every molecule in the gas moves with the same speed, v_{rms} .

- (A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E6. The energy needed to turn 0.200 kg of ice at -10°C into water at 20°C is ($c_{\text{ice}} = 2100$, $c_{\text{water}} = 4190 \text{ J kg}^{-1}\text{K}^{-1}$, $L_f = 3.34 \times 10^5 \text{ J/kg}$)

- (A) 21.0 kJ (B) 66.8 kJ (C) 83.6 kJ (D) 175.5 kJ (E) 87.8 kJ

E7. Water of mass 0.30 kg at 80°C is mixed with 0.50 kg of water at 20°C in an insulated container. The final temperature is

- (A) 35.0°C
- (B) 42.5°C
- (C) 50.0°C
- (D) 57.5°C
- (E) 60.0°C

E8. 0.050 kg of steam at 100°C is passed into 0.50 kg of water at 20°C in an insulated container ($L_v = 2.26 \times 10^6 \text{ J/kg}$). The final temperature is

- (A) 27.3°C
- (B) 60.0°C
- (C) 76.3°C
- (D) 88.0°C
- (E) 100.0°C

E9. Which of the following statements about phases of matter is/are true?

- I. At the triple point the solid, liquid and gas phases of a substance coexist in equilibrium.
- II. Above the critical temperature a gas can be liquefied by applying enough pressure alone.
- III. The melting curve of water has a negative slope, so pressure can melt ice at a fixed temperature just below 0°C .

(A) I only (B) I and II only (C) I and III only (D) II and III only (E) I, II and III

E10. The density of dry air ($M = 29.0\text{ g/mol}$) at $1.013 \times 10^5\text{ Pa}$ and 20°C is

- (A) 1.21 kg/m^3
- (B) 0.0121 kg/m^3
- (C) 1.29 kg/m^3
- (D) 2.41 kg/m^3
- (E) 1206 kg/m^3

Answers

- B1 (a) $p_1 = 321.3\text{ kPa absolute}$; $p_2 = 321.3 \times 323.15/283.15 = 366.7\text{ kPa}$; gauge 265 kPa .
 (b) $\sqrt{323.15/283.15} = 1.068$; $+6.8\%$. (c) The gas law needs absolute pressure; the gauge ratio would give $220 \times 1.141 = 251\text{ kPa}$
- B2 $V = 50\text{ m}^3$; $n = pV/RT = 2.08 \times 10^3\text{ mol}$; $N = 1.25 \times 10^{27}$; $m = nM = 59.9\text{ kg}$ (about 60 kg)
- B3 air 1.18 kg/m^3 ; helium 0.163 kg/m^3 ; the helium weighs 0.139 of the displaced air ($= M_{\text{He}}/M_{\text{air}}$)
- B4 $V_2 = 2.0 \times 3.0 \times (293.15/283.15) = 6.2\text{ cm}^3$
- B5 (b) $pV = 800, 1000, 800\text{ J}$; b is hottest; a and c are tied (same isotherm). (c) $nR = 3.326\text{ J/K}$; $T_a = T_c = 241\text{ K}$, $T_b = 301\text{ K}$
- B6 stages: 2.10×10^4 (ice), 1.67×10^5 (melt), 2.09×10^5 (water), 1.128×10^6 (boil), 2.01×10^4 (steam); total $1.55 \times 10^6\text{ J}$; boiling takes 73% ; $t = 773\text{ s}$ (12.9 min)
- B7 $m(L_f + c \times 4.0) = 0.250(4186)(18)$, so $m = 18\,837/350\,744 = 0.054\text{ kg}$ (54 g)
- B8 Condensing the steam gives $2.256 \times 10^5\text{ J}$, more than the $1.67 \times 10^5\text{ J}$ needed to melt the ice, so all the ice melts and the final state is liquid. $0.100(2.256 \times 10^6) + 0.100(4186)(100 - T_f) = 0.500(3.34 \times 10^5) + 0.500(4186)T_f \Rightarrow T_f = 40.0^\circ\text{C}$
- B9 (a) No: 400 Pa is below the triple-point pressure; the ice sublimates directly to vapour.
 (b) Lower pressure lowers the boiling temperature. (c) The higher pressure raises the boiling temperature, so the food is hotter
- B10 (a) $v_{\text{rms}} = 484\text{ m/s}$; (b) 1.92 km/s ; both $K = 6.21 \times 10^{-21}\text{ J}$; factor $3.98 = \sqrt{32.0/2.02}$ (same K , smaller mass)
- B11 $T = Mv^2/3R = 1.12 \times 10^3\text{ K}$; $v \propto \sqrt{T}$, so T must increase by a factor of 4
- B12 $N/V = p/kT = 2.46 \times 10^{25}\text{ m}^{-3}$; spacing $= 3.4\text{ nm}$, about 10 molecular diameters, so the molecules occupy a tiny fraction of the volume and the point-like assumption is reasonable at this density
- B13 (a) $p_1 = nRT_1/V = 2.50 \times 10^5\text{ Pa}$. (b) $v \propto \sqrt{T}$, so $T_2 = 4T_1 = 1200\text{ K}$ (927°C); $p_2 = 4p_1 = 1.00 \times 10^6\text{ Pa}$. (c) $K_1 = \frac{3}{2}nRT_1 = 7.49\text{ kJ}$; $K_2 = 29.9\text{ kJ}$; ratio 4. (d) $p = \frac{2}{3}K_{\text{tot}}/V$ at fixed V : $\frac{2}{3}(22.46 \times 10^3)/0.0200 = 7.49 \times 10^5\text{ Pa} = p_2 - p_1$

Exam-style practice: answer key and brief solutions

Key: E1 D E2 B E3 D E4 D E5 B E6 E E7 B E8 C E9 C E10 A

E1 (D). $pV = nRT$ with $T = 300$ K (kelvin, not 27): $V = \frac{(2.5)(8.314)(300)}{1.013 \times 10^5} = 6.16 \times 10^{-2} \text{ m}^3 = 61.6 \text{ L}$. The value 56.0 L uses the molar volume at 0°C , which is too small at 27°C .

E2 (B). $\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$ (fixed n , temperatures in kelvin): $p_2 = p_1 \frac{V_1 T_2}{V_2 T_1} = (1.00)(0.667)(1.20) = 0.80 \text{ atm}$. Prediction: the volume rises by 50% and the temperature by 20%, so the pressure must fall.

E3 (D). $v_{\text{rms}} = \sqrt{3RT/M} = \sqrt{3(8.314)(300)/0.0280} = 517 \text{ m/s}$ (use M in kg/mol). The mean speed $\sqrt{8RT/\pi M} = 476 \text{ m/s}$ is a different average.

E4 (D). $v_{\text{rms}} \propto \sqrt{T}$, so doubling v_{rms} requires $T \rightarrow 4T = 1200 \text{ K}$. Doubling the temperature raises the speed only by $\sqrt{2}$.

E5 (B). I and II are true: $K = \frac{3}{2}k_B T$ does not depend on the mass, so $v_{\text{rms}} = \sqrt{3k_B T/m}$ is larger for smaller m . III is false: the speeds follow a distribution. Answer: I and II only.

E6 (E). Three stages: warm the ice, $mc\Delta T = 4.2 \text{ kJ}$; melt it, $mL_f = 66.8 \text{ kJ}$; warm the water, $mc\Delta T = 16.8 \text{ kJ}$. Total 87.8 kJ. The melting stage dominates; leaving it out gives 21.0 kJ.

E7 (B). Heat lost = heat gained: $0.30(80 - T_f) = 0.50(T_f - 20)$, so $T_f = \frac{(0.30)(80) + (0.50)(20)}{0.80} = 42.5^\circ\text{C}$. It lies closer to 20°C because the cold water has the larger mass; the simple average 50°C ignores the masses.

E8 (C). Heat released by the steam = heat gained by the cold water: $m_s L_v + m_s c(100 - T_f) = m_w c(T_f - 20)$, so $T_f = 76.3^\circ\text{C}$. Check: $113 \text{ kJ} + 0.05(4190)(23.7) = 118 \text{ kJ} = 0.5(4190)(56.3)$. Leaving out the latent heat gives 27.3°C , far too low.

E9 (C). I is true. II is false: above the critical temperature no pressure produces a distinct liquid. III is true: ice is less dense than water, so raising the pressure favours the liquid. Answer: I and III only.

E10 (A). $\rho = \frac{m}{V} = \frac{pM}{RT} = \frac{(1.013 \times 10^5)(0.0290)}{(8.314)(293.15)} = 1.21 \text{ kg/m}^3$. The value 1.29 is the density at 0°C , and 1206 results from using M in g/mol.